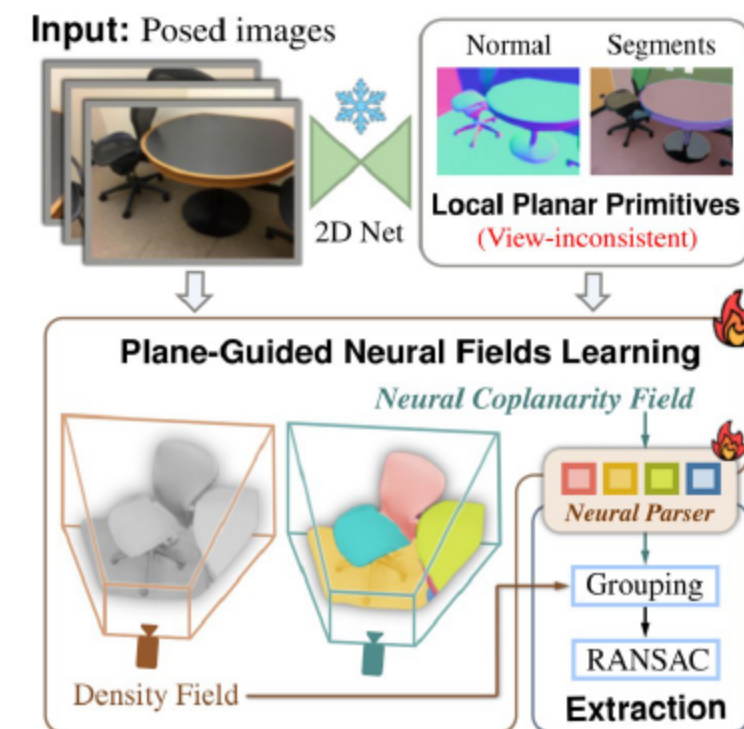


# NeuralPlane: Structured 3D Reconstruction in Planar Primitives

Hanqiao Ye, Yuzhou Liu, Yangdong Liu, Shuhan Shen

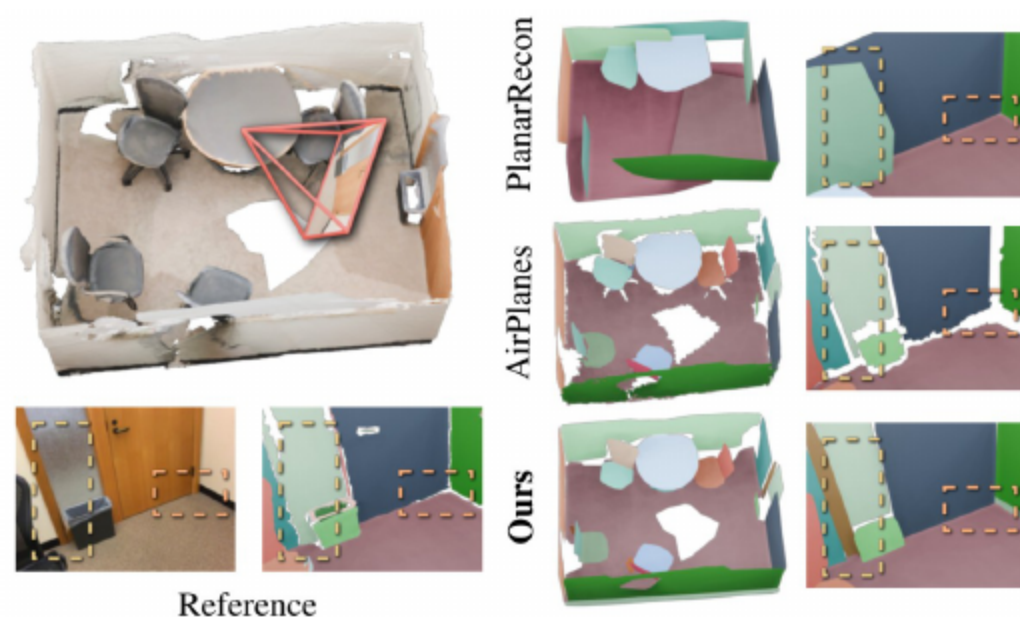
## 01 Dense Geometry to Structured Plane Maps



- Dense reconstructions from MVS or meshes are often **noisy and incomplete**; planar primitives are compact and semantic for man-made scenes
- Conventional geometry plus RANSAC pipelines decouple geometry and plane estimation and rely on high-quality inputs
- There is a need for a unified neural approach coupling geometry and semantics for consistent, clean plane maps across views

[Schonberger et al., 2016; Watson et al., 2024; Xie et al., 2022]  
[ARKit, 2024; ARCore, 2024]

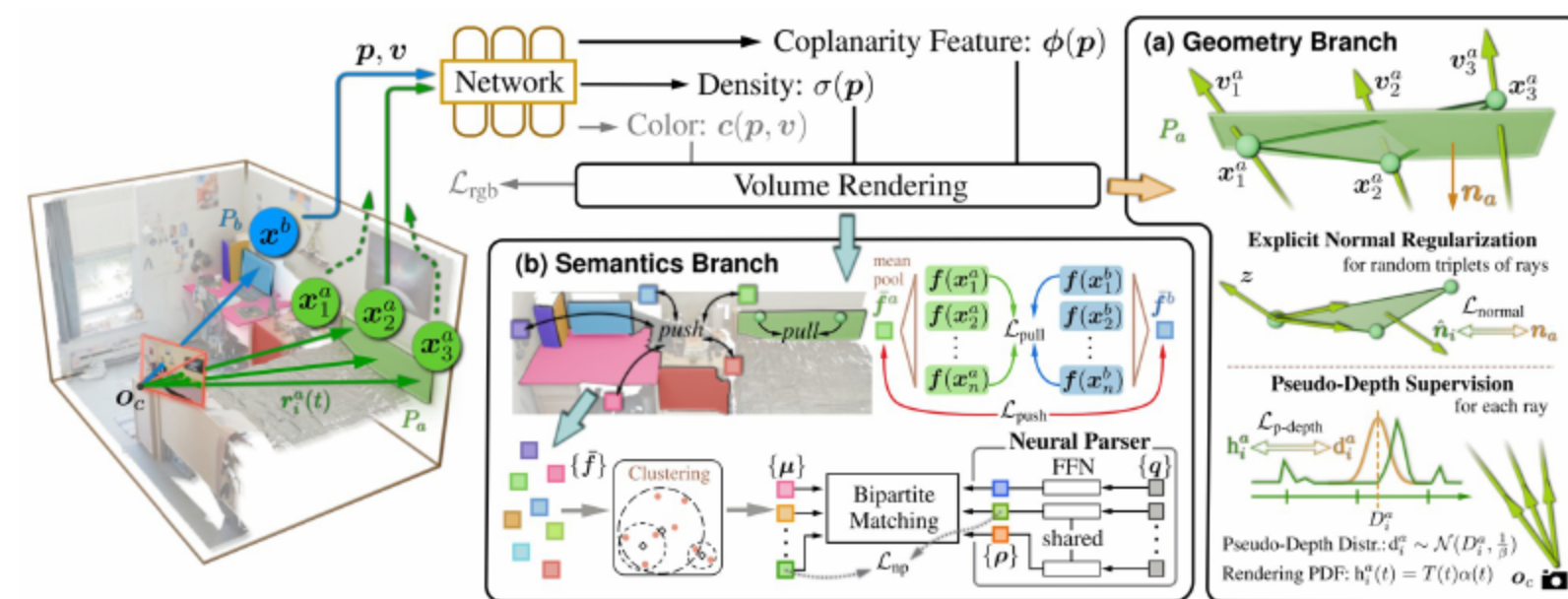
## 02 NeuralPlane Overview and Contributions



- Self-prompted initialization:** training-free 2D plane segments become Local Planar Primitives to initialize a plane-guided neural field
- Neural Coplanarity Field:** self-supervised contrastive features that capture coplanarity and semantics for robust separation
- Neural Parser:** query-based prototypes enabling plane-instance separation and adjustable level of detail

[Ye et al., 2023; Xue et al., 2024; Li et al., 2024]  
[Kim et al., 2024]

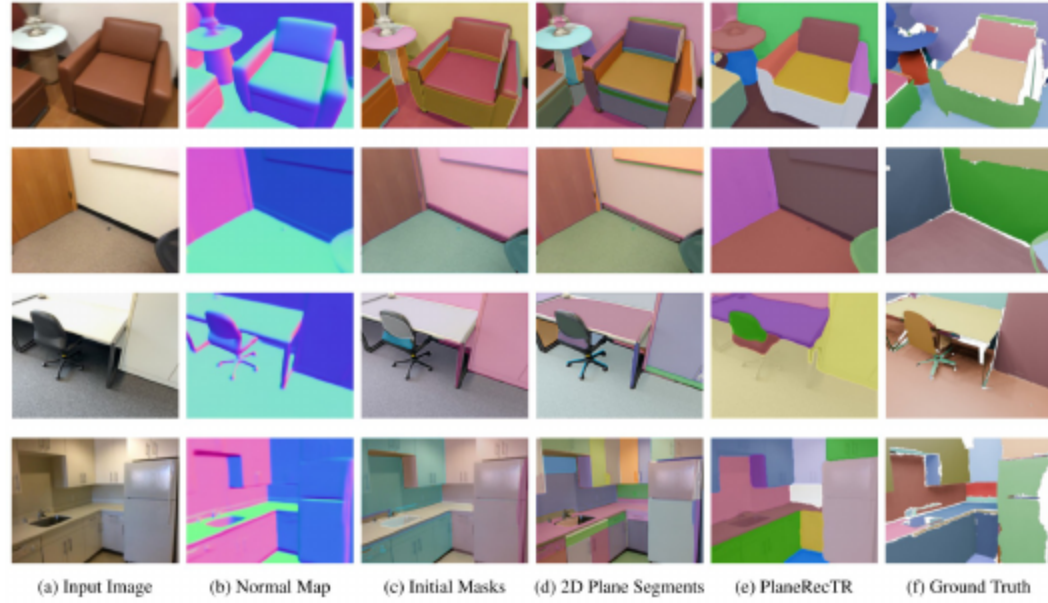
## 03 Pipeline at a Glance



- Two-branch training: a geometry branch for accurate plane localization and a semantics branch for coplanarity
- Local Planar Primitives guide density with normal and pseudo-depth regularizers; NCF learns contrastive plane features
- Extraction: prototype grouping via Neural Parser, then RANSAC fitting, then triangulated plane meshes



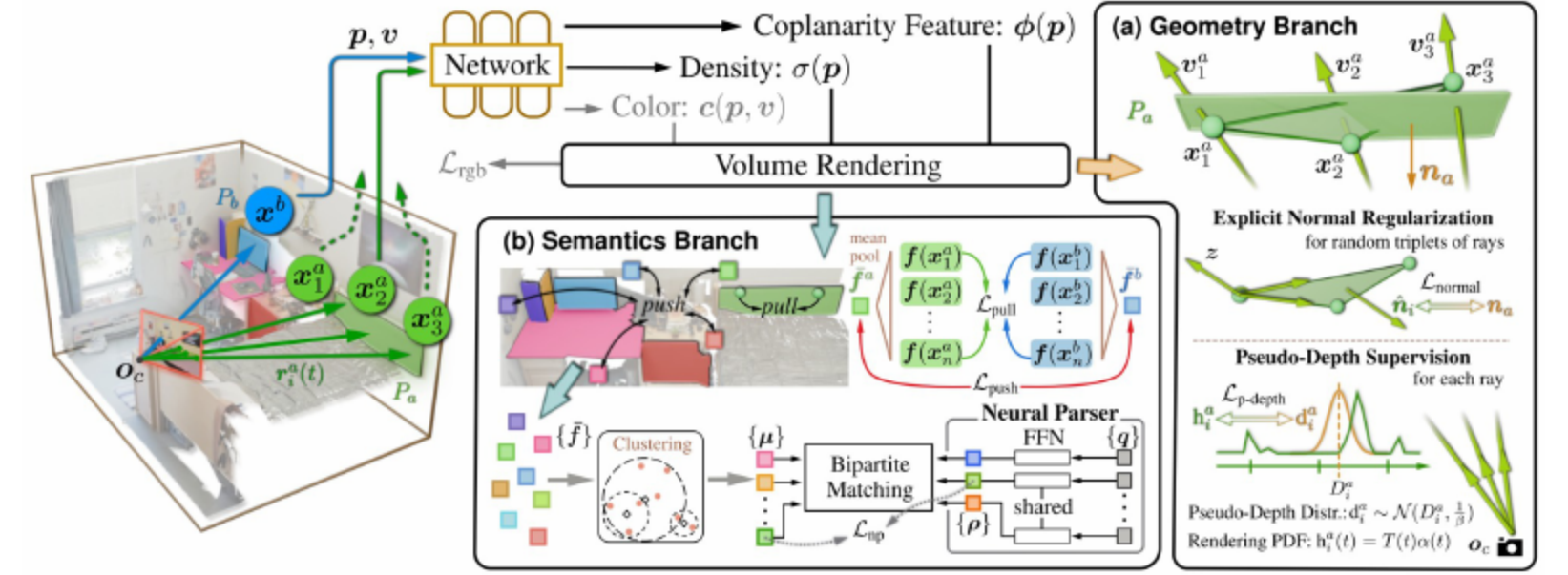
## 04 Local Planar Primitives: Monocular Segments



- Monocular normal clustering with K-means, followed by SAM over-segmentation to enforce geometric continuity
- Initialize plane parameters: the normal from averaged predicted normals; the offset from SfM keypoints weighted by reprojection error
- Annotation-free 2D cues provide coarse yet **effective** guidance across views despite cross-view inconsistency

[Bae et al., 2021; Kirillov et al., 2023; Lloyd, 1982]  
[Mazur et al., 2024]

## 05 Geometry Branch: Plane-Guided Regularization



- Normal loss aligns NeRF surface normals, estimated from ray triplets, to plane normals for smooth planar regions
- Pseudo-depth supervision with uncertainty matches the rendering probability to a target around the estimated depth
- Joint refinement lets gradients update plane parameters during training for **improved** localization

[Peng et al., 2022; Zhang et al., 2020]

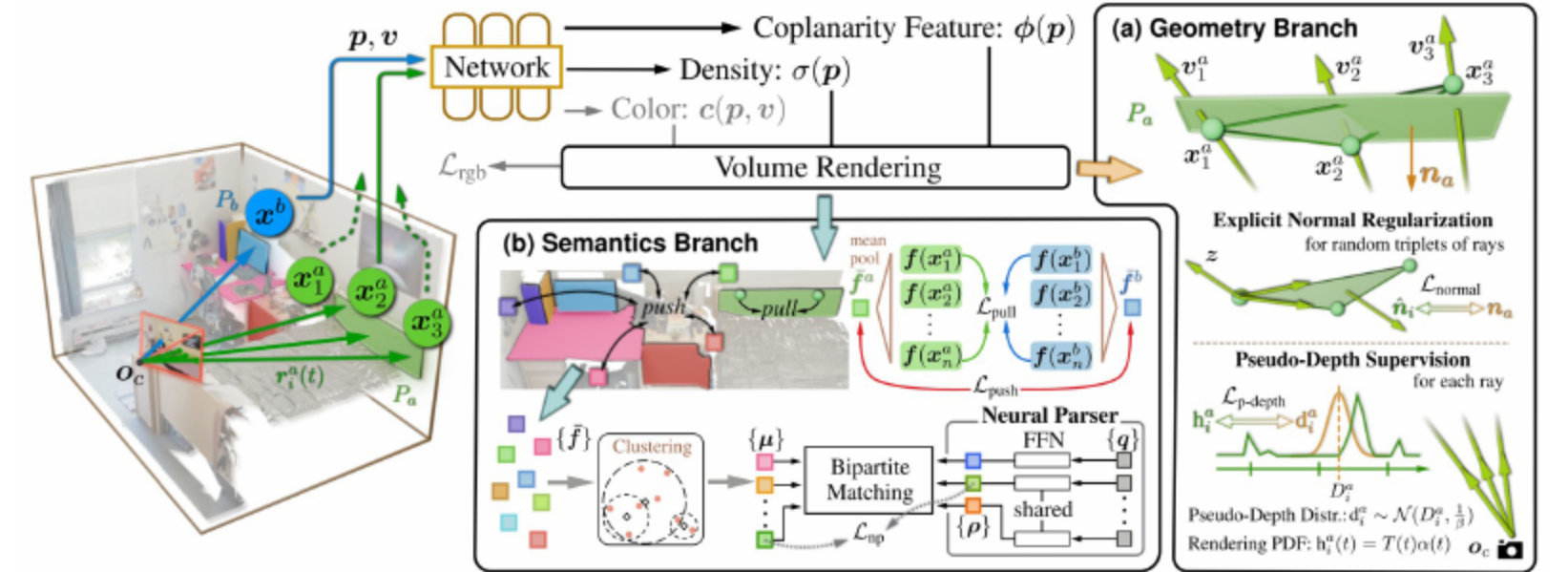
## 06 Semantics Branch: Neural Coplanarity Field (NCF)



- Define a feature field over 3D points and render a coplanarity feature by weighting features along each ray
- Contrastive learning uses a pull loss for rays within the same primitive and a push loss with geometry-aware negatives
- Learns semantics-aware planar boundaries and cross-view coplanarity consistency for **clean** separation

[Chopra et al., 2005; Hastie et al., 2009]

## 07 Neural Parser and Plane Extraction



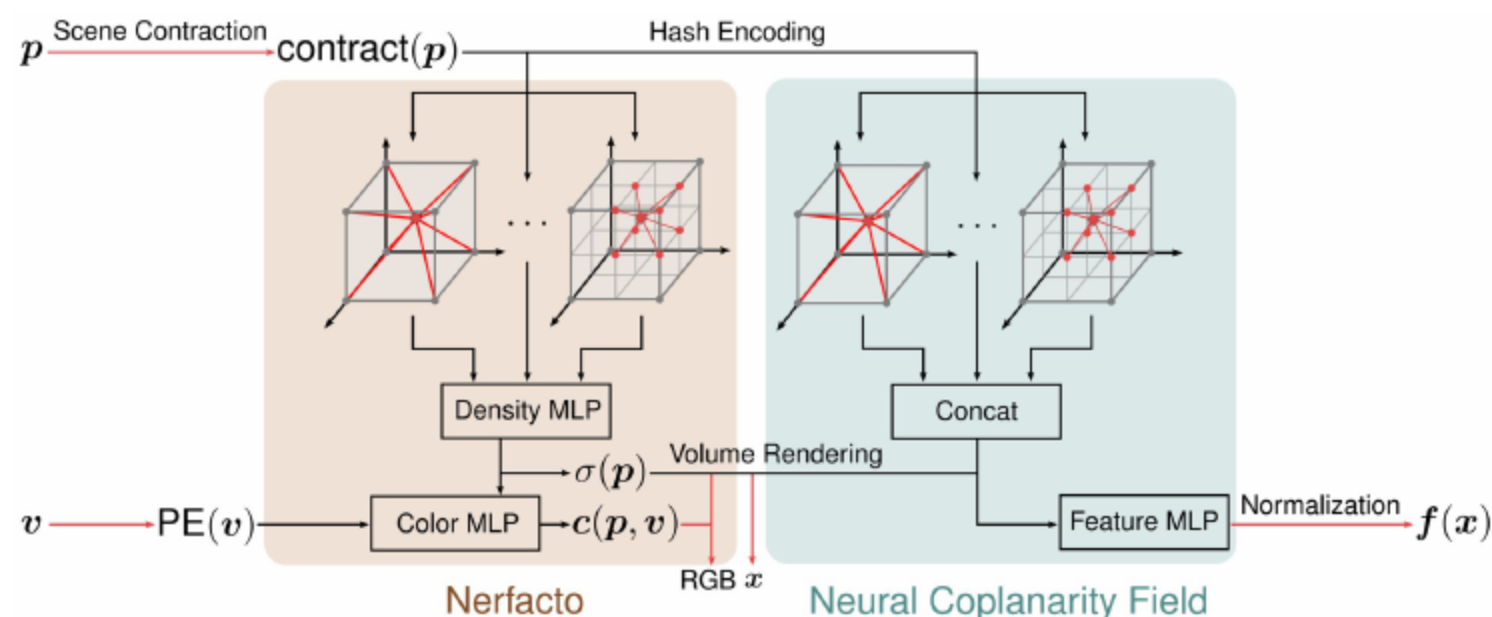
- Neural Parser: a query-based feed-forward network predicts a fixed number of prototypes as instance centroids
- During training, DBSCAN centroids are matched to prototypes via Hungarian assignment with separation and assignment losses
- Extraction groups by nearest prototype, fits planes with RANSAC, and triangulates inliers into meshes

[Ester et al., 1996; Jorner & Vongnont, 1987]



NeRF represents density and color. For a camera ray with sampled points along its path, the expected termination point is the weighted sum of sample locations, where the weights approximate the rendering probability. Opacity at each sample depends on the local density and the sampling step.

[Mildenhall et al., 2020]



- Baselines: PlanarRecon and geometry plus RANSAC using MVS or neural surfaces such as ManhattanSDF, NeuRIS, MonoSDF, and ObjectSDF++
- Use AirPlanes and Sequential RANSAC for plane fitting on baseline geometries with official pretrained models
- Implementation on Nerfstudio and Nerfacto with a fixed set of hyperparameters across scenes

[Tancik et al., 2023; Xie et al., 2022; Sayed et al., 2022]  
[Stier et al., 2023; Wang et al., 2021]

|           | Scene ID   | Type            | # of Frames | Down. Rate | Area (m <sup>2</sup> ) |
|-----------|------------|-----------------|-------------|------------|------------------------|
| ScanNetv2 | 0084_00    | bathroom        | 247         | 12.5 %     | 42.68                  |
|           | 0164_03    | kitchen         | 140         | 12.5 %     | 28.97                  |
|           | 0217_00    | bedroom         | 156         | 12.5 %     | 40.65                  |
|           | 0316_00    | lounge          | 97          | 12.5 %     | 26.56                  |
|           | 0356_00    | bedroom         | 170         | 12.5 %     | 27.77                  |
|           | 0427_00    | conference room | 165         | 12.5 %     | 24.11                  |
|           | 0488_01    | kitchen         | 137         | 12.5 %     | 38.53                  |
|           | 0568_00    | lounge          | 207         | 12.5 %     | 71.49                  |
| ScanNet++ | f6659a3107 | conference room | 181         | 1.8 %      | 133.38                 |
|           | 31a2c91c43 | bathroom        | 184         | 2.4 %      | 31.72                  |
|           | 7bc286c1b6 | bathroom        | 150         | 3.3 %      | 25.13                  |
|           | 303745abc7 | office          | 279         | 4.4 %      | 63.43                  |

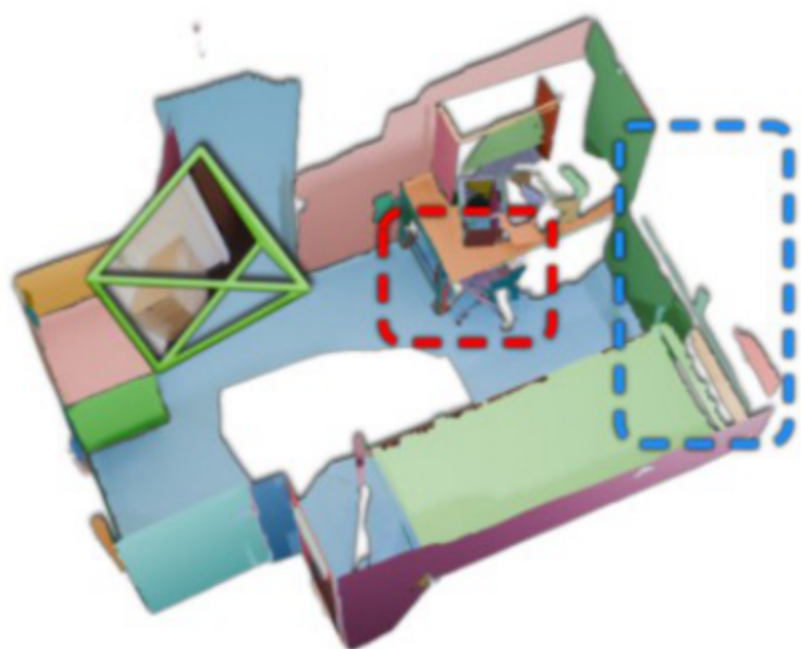
- Datasets: eight ScanNetv2 and four ScanNet++ indoor scenes with diverse and often low-texture layouts
- Geometry metrics: accuracy, completeness, Chamfer distance, and precision, recall, F-score at five centimeters
- Segmentation metrics: rand index, variation of information, and segmentation covering

[Dai et al., 2017; Yeshwanth et al., 2023; Bozic et al., 2021]  
[Arbeláez et al., 2011]

| Method                                        | Geometry    |             |              |             |             |             | Segmentation |       |              |
|-----------------------------------------------|-------------|-------------|--------------|-------------|-------------|-------------|--------------|-------|--------------|
|                                               | Accu. ↓     | Comp. ↓     | Chamfer ↓    | Prec. ↑     | Recall ↑    | F-score ↑   | RI ↑         | VOI ↓ | SC ↑         |
| COLMAP (Schönberger et al., 2016)             | 18.35       | 13.40       | <b>15.88</b> | 40.5        | 42.4        | <b>41.2</b> | -            | -     | -            |
| SimpleRecon (Sayed et al., 2022)              | 6.74        | 5.55        | <b>6.15</b>  | 59.2        | 58.8        | <b>59.0</b> | -            | -     | -            |
| FineRecon (Stier et al., 2023)                | 6.12        | 4.19        | <b>5.16</b>  | 68.6        | 72.9        | <b>70.6</b> | -            | -     | -            |
| ManhattanSDF (Guo et al., 2022)               | 9.95        | 7.51        | <b>8.73</b>  | 50.4        | 50.9        | <b>50.6</b> | -            | -     | -            |
| NeuRIS (Wang et al., 2022)                    | 10.23       | 5.69        | <b>7.96</b>  | 62.0        | 64.8        | <b>63.2</b> | -            | -     | -            |
| MonoSDF <sup>†</sup> (Yu et al., 2022b)       | 5.28        | 5.07        | <b>5.18</b>  | 69.7        | 69.7        | <b>69.7</b> | -            | -     | -            |
| ObjectSDF++ <sup>†</sup> (Wu et al., 2023a)   | 8.82        | 5.25        | <b>7.03</b>  | 54.9        | 68.9        | <b>60.9</b> | -            | -     | -            |
| COLMAP                                        | 19.32       | 13.37       | <b>16.34</b> | 40.6        | 40.9        | <b>40.6</b> | 0.928        | 3.91  | 0.154        |
| SimpleRecon                                   | 6.24        | 6.00        | <b>6.12</b>  | 57.7        | 52.6        | <b>54.9</b> | 0.949        | 2.65  | 0.272        |
| FineRecon                                     | 5.20        | 5.65        | <b>5.43</b>  | 69.1        | 64.9        | <b>66.7</b> | 0.941        | 2.56  | 0.276        |
| ManhattanSDF +Seq.RANSAC                      | 9.37        | 8.70        | <b>9.04</b>  | 50.6        | 51.0        | <b>50.8</b> | 0.930        | 2.82  | 0.281        |
| NeuRIS                                        | 9.87        | 6.35        | <b>8.11</b>  | 59.6        | 59.3        | <b>59.3</b> | 0.945        | 2.57  | 0.293        |
| MonoSDF                                       | 5.91        | 5.43        | <b>5.67</b>  | 65.9        | 66.1        | <b>65.9</b> | 0.945        | 2.38  | 0.333        |
| ObjectSDF++ <sup>†</sup>                      | 8.41        | <b>5.35</b> | <b>6.88</b>  | 58.0        | <b>68.8</b> | <b>62.8</b> | 0.952        | 2.32  | 0.334        |
| SimpleRecon                                   | 5.43        | 6.60        | <b>6.01</b>  | 59.2        | 51.5        | <b>55.1</b> | 0.944        | 2.51  | 0.341        |
| FineRecon                                     | <b>4.93</b> | 5.95        | <b>5.44</b>  | <b>70.8</b> | 62.2        | <b>66.2</b> | 0.947        | 2.43  | 0.310        |
| ManhattanSDF +AirPlanes (Watson et al., 2024) | 9.46        | 9.30        | <b>9.38</b>  | 52.7        | 50.2        | <b>51.3</b> | 0.940        | 2.61  | 0.315        |
| NeuRIS                                        | 6.05        | 7.38        | <b>6.71</b>  | 66.2        | 56.7        | <b>61.0</b> | 0.943        | 2.55  | 0.291        |
| MonoSDF                                       | <b>4.57</b> | 6.33        | <b>5.45</b>  | <b>72.2</b> | 62.0        | <b>66.6</b> | 0.948        | 2.38  | <b>0.346</b> |
| PlanarRecon (Xie et al., 2022)                | 7.74        | 11.85       | <b>9.80</b>  | 55.2        | 44.3        | <b>49.0</b> | 0.909        | 3.27  | 0.265        |
| NeuralPlane@PlaneRecTR                        | 5.38        | <b>4.65</b> | <b>5.02</b>  | 67.6        | <b>70.0</b> | <b>68.7</b> | 0.949        | 2.37  | 0.364        |
| NeuralPlane                                   | <b>4.92</b> | <b>4.27</b> | <b>4.59</b>  | <b>70.5</b> | <b>71.9</b> | <b>71.2</b> | 0.955        | 2.25  | 0.376        |

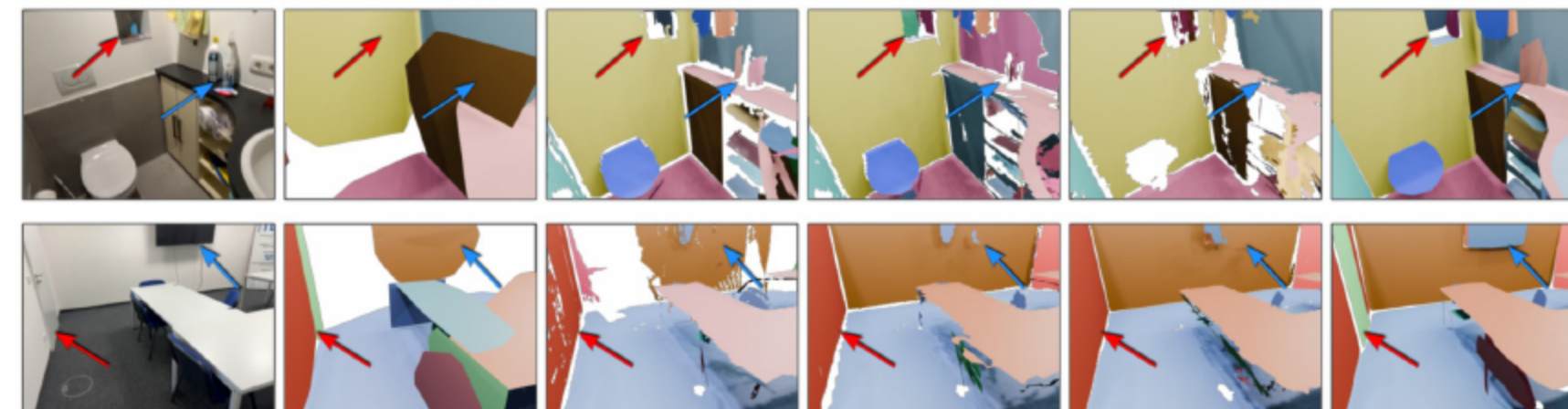
- NeuralPlane reaches **state-of-the-art** geometry with lower Chamfer and higher F-score and strong segmentation with higher rand index and covering and lower variation of information
- AirPlanes variants show high precision but **reduced recall**; NeuralPlane balances both
- Outperforms ObjectSDF++ plus RANSAC in segmentation without ground truth instance supervision

## 12 Qualitative Results on ScanNetv2



- Crisp planar structures with fine details and consistent semantics across views
- Avoids over-merging adjacent planes compared to geometry plus RANSAC baselines
- Preserves scene layout while segmenting semantically distinct coplanar regions

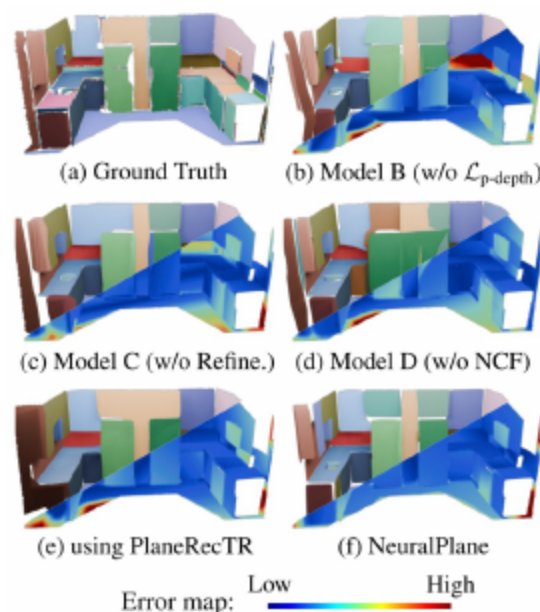
## 13 Results on ScanNet++



Reference View PlanarRecon NeuRIS+AirPlanes NeuRIS+RANSAC MonoSDF+RANSAC NeuralPlane

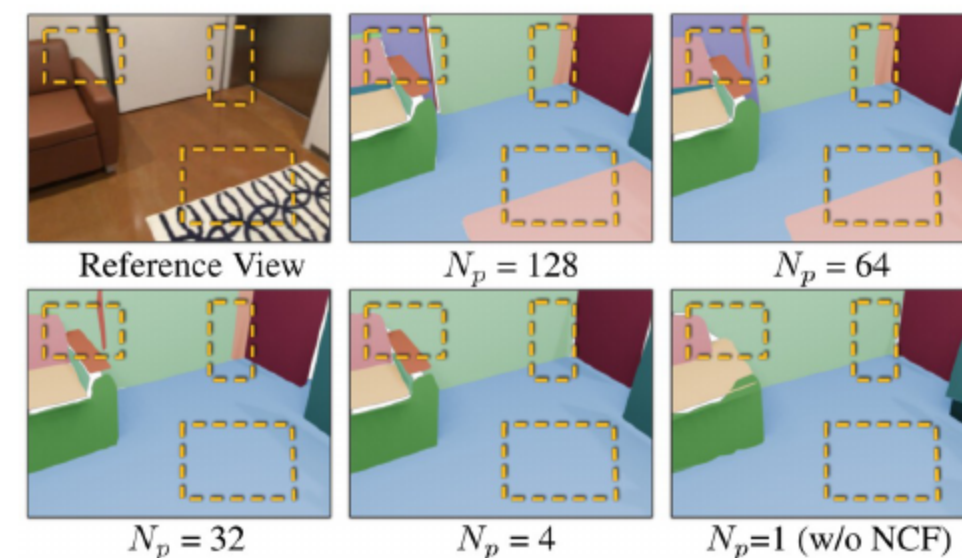
- Competitive geometry on low-texture scenes and consistently **strong** segmentation quality
- Robust to sparse or misleading SfM keypoints via implicit fusion of local planar primitives in the neural field
- Efficient training from density representation with large reported speedups over neural surface baselines

## 14 Component Analysis



- Removing geometric regularizers **degrades** reconstruction quality in planar regions
- Removing semantics causes **conflicts** at boundaries and poorer instance separation
- Neural Parser outperforms post-clustering for instance separation and final plane quality; the full model yields the **best** geometry and segmentation

## 15 Semantic Prototypes ( $N_p$ )



(b) Different Levels of Detail

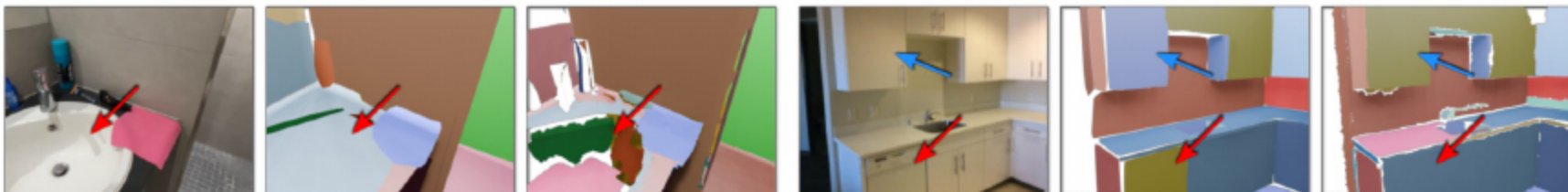
- The number of prototypes controls semantic level of detail for plane instances
- Performance is best around thirty-two prototypes; too many can cause **over-segmentation**
- Similar trends across datasets; geometry and segmentation both **degrade** when the number is excessive





(a) Missing structures

(b) Under-segmentation



(c) Non-planar surfaces

(d) Over-segmentation

- Missed planes near the optical axis, under or over-segmentation, and non-planar regions forced into planes are **failure modes**
- Future work: stronger cross-view consistency cues, better geometry-aware negatives, and adaptive prototype count
- Reduce dependence on SfM via learned depth priors and robust keypoint selection for **improved stability**

**THANKS!**